TEXORA · LECTURE SHEET

Static Electricity Formation

Charge that will not leave: triboelectric generation at every contact and separation, the triboelectric series, exponential decay governed by resistance, the processing chaos and hazards static causes in mills and garments — and the layered defences, from humidity to conductive fibres.

TE-3209 · TEXTILE PHYSICS · TOPIC 10 · LEVEL 3 TERM 2

1. Generation — contact, separation, charge

When two materials touch, electrons (and sometimes ions) transfer across the interface to equalise surface energy states; separate them quickly and each carries away its share — one positive, one negative. Rubbing is only vigorous repeated contact–separation; generation also occurs at fibre–roller contacts, in fibre opening and carding, at fabric peeling from drums, in pneumatic transport and even in liquid flow (streaming electrification in solvent lines). The amount retained is a race between generation and leakage: charge accumulates until the leak rate (set by Topic 9's resistance) balances generation, or until the field breaks down in a spark.

Which partner goes positive is systematised by the **triboelectric series** — materials rubbed with anything below them charge positive:

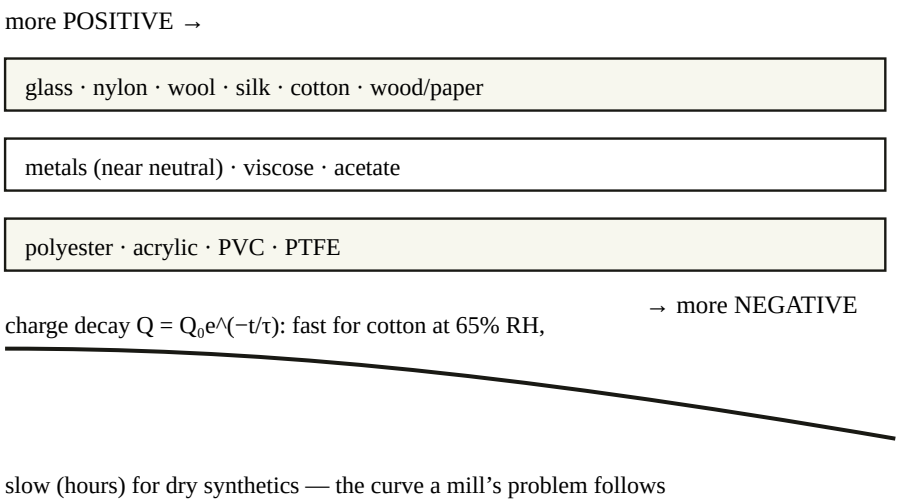


Figure 1 — Top: the triboelectric series (simplified): rub any pair and the higher one charges positive. Bottom: exponential charge decay — the time constant is set by the material's resistance.

The series explains garment cling pairings (a polyester lining against a wool dress charges hard because the pair sits far apart) and predicts polarity on rollers and guides. It is empirical — surface contamination can move a material several places — so it is used for ranking and sign, not precise voltages.

2. Decay — the exponential leak

Isolated charge leaks away through and over the material with exponential decay:

$$Q = Q_0\, e^{(-t/\tau)}, \quad \tau = RC \text{ — quoted practically as the half-life } t_{½} = 0.693\tau$$

WORKED EXAMPLE — HALF-LIFE ARITHMETIC

A fabric's charge half-life at working humidity is 2 s. Starting from 100% charge: after 2 s → 50%; 4 s → 25%; 10 s (five half-lives) → $(½)^5 = 3.1\%$ — effectively gone within a machine's transit. Now dry the air until the same fabric's half-life stretches to 20 minutes: after 10 s, ≈ 99.4% remains — the web reaches the next roller fully charged, and every downstream symptom (clinging, sparking, dust pickup) is scheduled. One number, $t_{½}$ at the actual RH, predicts the whole behaviour; specifications for antistatic materials are written in exactly these terms.

3. Consequences

- **Processing:** fibre fly and lap-ups in carding, ballooning and snarling yarns, fabric clinging to rollers and folding tables, shocks to operators, dust and lint attraction that soils goods and fouls machines, erratic behaviour of webs at high speed (nonwovens, film-like fabrics).
- **Hazards:** spark discharge in solvent vapour or dust clouds (fire/explosion risk in coating and flock plants), damage to electronics during garment/finishing work near devices, painful (rarely dangerous) personnel shocks at kilovolt potentials but tiny energies.
- **In wear:** garment cling, crackling, dust pickup on dark synthetics; in cleanrooms and operating theatres, static is a contamination and safety specification, not a nuisance — hence conductive-fibre workwear standards.

4. Control — attack generation, resistance, or both

METHOD	PRINCIPLE AND LIMITS
Raise humidity (65–70% RH)	Surface water film slashes resistance of hydrophilic materials (Topic 9); cheap and broad — but useless for hydrophobics and uncomfortable/unaffordable at extremes

Antistatic finishes	Hygroscopic/ionic coatings lower surface resistance; durability to washing varies; dry-air weakness (Topic 9)
Conductive fibres	Carbon-core, metal-coated or stainless fibres at 0.5–2% bleed charge continuously and wash-proof — the engineering solution for workwear, carpets, conveyor belts
Ionisers / static bars	Corona points flood the web with both polarities of air ions, neutralising charge at the point of need — standard on winders and stenters; needs placement at separation points
Earthing & material pairing	Grounded rollers and machine frames drain conductive paths; choosing guide/clothing materials near the fibre in the triboelectric series reduces generation at source
Process design	Lower speeds at separation points, larger wrap/roller radii, avoiding unnecessary contact-separation cycles

5. Key points

- Static is contact–separation charge transfer racing against leakage; the triboelectric series predicts sign and severity of pairings.
- Decay is exponential with half-life set by resistance and humidity: 2 s half-life clears charge in ~10 s; a dry-day 20-minute half-life clears nothing.
- Consequences scale from nuisance (cling, dust) to plant hazards (sparks in vapour) — classify before spending.
- Defences layer: humidity, antistatic finishes, conductive fibres, ionisers, earthing, gentler process geometry.
- Polyester’s position in the series plus its dryness (Topics 2, 9) make it the static champion of apparel fibres.

6. Practice

Answer in your own words; brief model answers follow.

1. A web has half-life 5 s at the plant’s humidity. What fraction of charge remains after 15 s? After 1 s?
2. Using the series, predict the sign on (a) a nylon brush roller working a polyester web, (b) a cotton fabric peeled from a PVC belt.
3. Static complaints peak in winter in a heated mill although the fibre mix never changes. Explain in two linked steps.

4. An antistatic carpet specification demands charge decay to 10% within 2 s at 25% RH. What half-life must the carpet achieve?
5. Why are ionising bars placed immediately after separation points (e.g., just after a roller nip) rather than anywhere convenient along the line?
6. A solvent-coating line asks whether antistatic finish alone is sufficient protection. Answer with the hazard logic and the layered measures you would insist on.

Answer key

1. After 15 s = 3 half-lives → (½)³ = 12.5%. After 1 s = 0.2 half-lives → 2^(−0.2) = 0.87 → 87% remains.
2. (a) Nylon sits well above polyester: nylon charges positive, the web negative. (b) Cotton is above PVC: cotton positive, belt negative — and PVC's high resistance keeps its charge, worsening the peel problem.
3. Heating cold winter air drops its relative humidity drastically; lower RH raises fibre surface resistance (Topic 9), lengthening charge half-life, so the same generation now accumulates instead of leaking — the fibre mix is innocent; the air is guilty.
4. 10% remaining means (½)^n = 0.1 → n = $\log_2 10$ = 3.32 half-lives in 2 s → t½ ≤ 2/3.32 = 0.60 s at a hostile 25% RH — a specification only conductive-fibre constructions reliably meet; finishes alone would fail the dry condition.
5. Charge is created at the separation instant and begins attracting dust/reattaching immediately; neutralising ions must meet the web while charge is accessible and before it reaches the next contact — downstream placement lets charged material act (cling, spark, attract) in between and buries charge under wound layers where ions cannot reach.
6. No. In a flammable-vapour zone the requirement is that no incendive spark can occur: finishes are humidity-dependent and wear off. Insist on layered, independent controls: conductive/grounded plant and belting, conductive fibres or permanent antistats verified at low RH, ionisation at separation points, vapour extraction/ventilation and zoning — protection by systems, not by a coating.
